

From Labeling Strategies to Cross-Season Generalization: A Multimodal Deep Learning Framework for Tree-Level Prediction of Pomegranate Fruit Cracking

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Abstract

This study presents a multimodal deep learning framework for predicting on-tree pomegranate fruit cracking severity by integrating RGB imagery, thermal data, and meteorological data. The system combines YOLOv8 fruit detection with transfer learning and achieved 86.6% mAP50 and 54.1% mAP50-95, U-Net segmentation that used grab cut heuristic masks for training with 84.5% IoU, and a CNN+LSTM classifier that models spatial and temporal information from three consecutive time points. Class separation is supported through linear discriminant analysis (LDA). Three labeling strategies were evaluated. The original three-class labels performed weakest, with a best result of 68.8% accuracy and 55.7% agreement accuracy (AA; measuring consistency between north and south side predictions for the same tree). The fuzzy labeling approach, using Gaussian membership values, reached 0.060 MSE, 0.007 inter-side MSE (measuring consistency between north and south side fuzzy predictions for the same tree) and 76.7% composite accuracy (CA; a unified measure enabling a numeric comparison between hard-label and fuzzy-soft-label strategies). The binary scheme achieved the strongest performance within the 2024 season with 84.1% accuracy and 70.4% AA. Additionally, cross-season generalization evaluation was performed under the train-2024/test-2025 setup, in which the 2025 RGB images were first segmented using the U-Net model trained on 2024, and the resulting segmented images were used together with the thermal and environmental data in the same multimodal configuration. Under this setup, the original three-class labels degraded most strongly with 35% accuracy and 11.3% AA. The fuzzy labels achieved 0.068 MSE, 0.017 inter-side MSE and 68.0% CA. The binary scheme showed moderate robustness with 65.0% accuracy and 36.9% AA. Overall, while the binary labeling scheme provided the most accurate predictions, the gaussian fuzzy labeling scheme formulation generalized best across seasons.

Keywords: Pomegranate Fruit Cracking, Cracking Prediction, Deep Learning, Multimodal Sensing, Fruit Agriculture, Labeling Strategies, Cross Season Generalization.

1. Introduction

Fruit cracking is a major physiological disorder that reduces fruit quality, shelf life and market value. The peel of most fruits protects against mechanical injury, pathogens, and water loss, while allowing gas exchange. In apples and plums, the peel consists of a single exocarp, whereas in citrus and pomegranates it includes both the exocarp and mesocarp (Lufu et al., 2021). Cracking can occur during fruit growth, ripening, or even after harvest due to environmental stress (Opara, 1996). Surface cracks may deepen into the flesh, causing premature splitting, faster water loss and easier pathogen entry. Susceptibility depends on both internal factors fruit size, shape and skin properties (Emmons & Scott, 1998) and external influences such as temperature, humidity, soil moisture and fertilization, where fluctuations rather than single parameters are decisive (García-Luis et al., 1994; Peet, 1992). Orchard management, especially irrigation and nutrition, also affects cracking incidence (Khadivi-Khub, 2015). Studies comparing skin characteristics across species have clarified mechanical and anatomical traits linked to cracking resistance (Considine & Brown, 1981; Emmons & Scott, 1998; Knoche & Lang, 2017).

Pomegranate fruit ([Fig. 1](#)) has a thick, spongy white inner mesocarp and a smooth, cuticle-covered epidermal layer that together form the peel (Chang et al., 2019). Cracking begins as micro-fractures in the cuticle that deepen into the underlying tissues until the arils become exposed. While excess irrigation and rainfall promote cracking in many fruits, pomegranate behaves differently: reduced water availability and high temperatures during the rapid fruit-growth stage can increase cracking susceptibility (Joshi et al., 2021). As a major tropical and subtropical crop, pomegranate has seen expanding global production and trade (Singh et al., 2020), yet cracking losses may exceed 65 % in arid regions (Chandra et al., 2011; Malhotra et al., 1983; Prasad et al., 2003). Reported severity ranges from 18 % to over 60 % depending on cultivar (Singh et al., 2014), with economic losses typically 10-40 % and sometimes reaching 70 % (Pal et al., 2019). Seasonal effects are also evident, with higher cracking incidence in spring crops than in winter or rainy seasons (Udyanika, 1976). Overall, cracking can reduce the fruit's market value by about 50 % (El Sayed et al., 2014).



Fig. 1. Example photograph of cracked pomegranate fruit.

Deep learning (DL) has become central to modern agricultural prediction systems, enabling end-to-end modeling directly from different data sources such as images, weather, and soil measurements. By learning hierarchical representations through multiple nonlinear layers, DL models can capture complex spatial and temporal relationships within high-dimensional agricultural datasets (Benos et al., 2021; Lecun et al., 2015). Convolutional neural networks (CNNs) are widely used for spatial feature extraction from field imagery, while recurrent neural networks (RNNs) model temporal dependencies, allowing integration of image sequences and environmental variables (Mathew et al., 2021). These architectures have driven major progress in crop monitoring, stress detection and yield forecasting (Attri et al., 2023; Kamilaris & Prenafeta-Boldú, 2018). Prediction models using CNNs

and hybrid CNN-RNN architectures have achieved high accuracy for yield estimation across crops such as wheat, corn, and soybeans by effectively combining visual and environmental features (Khaki et al., 2021; Srivastava et al., 2022).

Real-world classification datasets are rarely noise-free, and noise can hinder analysis and the interpretation of learned models and decisions (Zhong et al., 2004; Zhu & Wu, 2004). In classification tasks, noise negatively affects performance, including accuracy and interpretability (Zhong et al., 2004; Zhu & Wu, 2004). In particular, class noise (labeling errors) occurs when examples are incorrectly labeled due to causes such as subjectivity in labeling or inadequacy of the information used to assign labels (Wu, 1996). Such noise can affect the intrinsic characteristics of a classification problem by altering class boundaries and class overlap and by creating misleading local clusters, which makes knowledge extraction more difficult and can spoil the learned models compared with models learned from clean data (Zhu & Wu, 2004). In deep learning-based agricultural applications, similar prediction tasks have been explored using different label formulation definitions. For example, papaya leaf disease classification has been evaluated using both binary (healthy vs. diseased) and multi-class disease labeling schemes on the same dataset (Kilic & Sağlam, 2025). Similarly, (Sunil et al., 2022) evaluated binary and multi-class disease classification using leaf images from cherry, grape, maize, pepper, potato, strawberry, and cardamom.

Recent studies have increasingly applied predictive modeling and sensor-based monitoring to mitigate fruit cracking. Early approaches focused on strengthening the peel or adjusting irrigation and fertilization, whereas recent work integrates data-driven prediction and environmental sensing. (Ehret et al., 2008) used artificial neural networks with environmental and crop variables such as temperature, humidity, CO₂, and fruit mass to predict cracking in pepper and tomato, achieving R² values of 0.71 and 0.70 for pepper and 0.70 and 0.64 for tomato at 1- and 4-week intervals. (Ranjan et al., 2022) combined thermal-RGB imagery with weather data to model fruit surface temperature in sweet cherries using stepwise multilinear regression, reaching R² = 0.80 (RMSE = 8.7%) and up to R² = 0.86 (RMSE = 3.5%) for specific cultivars. (Abekasis et al., 2024) integrated climate, management and spatial data from 250 citrus orchards using TSFRESH and H₂O-AutoML, achieving 76% classification accuracy and MAE of 4.78%. While these models highlight the potential of data-driven prediction, none have combined multimodal proximal sensing with deep learning to forecast cracking severity before visible symptoms, particularly in pomegranate trees.

Although the mechanical and environmental factors influencing cracking fruits are different from those in engineering materials, both domains share the need to predict failure processes that depend on multiple interacting variables. Insights gained from industrial applications, particularly in the integration of data-driven modeling and real-time monitoring, can inform the development of predictive frameworks for fruit cracking. In recent years, DL has improved crack prediction accuracy in engineering, with the Neural Network Incremental Learning (NNIL) model an adaptive neural network that updates its parameters incrementally during training reaching an MSE of 0.10 for aluminum specimens (Ma et al., 2021) and the Temporal Data-Attention Bidirectional Long Short-Term Memory (TDA-Bi-LSTM) model which combines bidirectional sequence learning with an attention mechanism showing the lowest MAE and RMSE across fatigue life intervals (Jian et al., 2025). These results demonstrate how multimodal data fusion with DL architectures enhances early crack detection (Yuan et al., 2024), a principle that can similarly be applied to biological cracking prediction in fruit crops.

Proximal sensing enables near-field data collection directly from the tree canopy using ground-based sensors (Alexopoulos et al., 2023). In this study, it was implemented through RGB and thermal imaging captured by a handheld vehicle, combined with environmental data. Together, these modalities formed a complementary multimodal sensing setup, where each modality provided distinct and non-overlapping information, improving stability and interpretability under varying orchard conditions (Khaleghi et al., 2013; Lahat et al., 2015). The main objective of this study is to evaluate multiple labeling schemes strategies multi-class, fuzzy, and binary to better represent uncertainty in cracking progression and to identify the most stable and interpretable label formulation for early prediction of tree pomegranate fruit cracking. A further objective is to assess model robustness and cross-season generalization across all labeling strategies by training the models on 2024 season data and evaluating their performance on an independent 2025 season dataset. These objectives are evaluated using the optimized multimodal CNN+LSTM model (Peleg., A., 2026). While previous multimodal proximal-sensing studies in agriculture have largely focused on post-occurrence analysis or short-term environmental responses, the present work demonstrates a generalizable framework capable of predicting fruit cracking before visible symptoms emerge.

2. Materials and methods

2.1. Dataset acquisition

Data collection (שגיאה! מקור ההפניה לא נמצא.) for this research was conducted using the TOMMY system (section 2.1.1) equipped with close range thermal and RGB sensors (specifications Table 2), together with a meteorological station that was manually activated at the beginning of each session to continuously record environmental data throughout the process. Data was collected by from August to November 2024 and 2025 near Tzora, Beit Shemesh, Israel ($31^{\circ}45'33.3''N$, $34^{\circ}57'20.5''E$), during the growth period of the Wonderful pomegranate cultivar (Fig. 2).



Fig. 2. The image shows where pomegranate tree data was acquired.

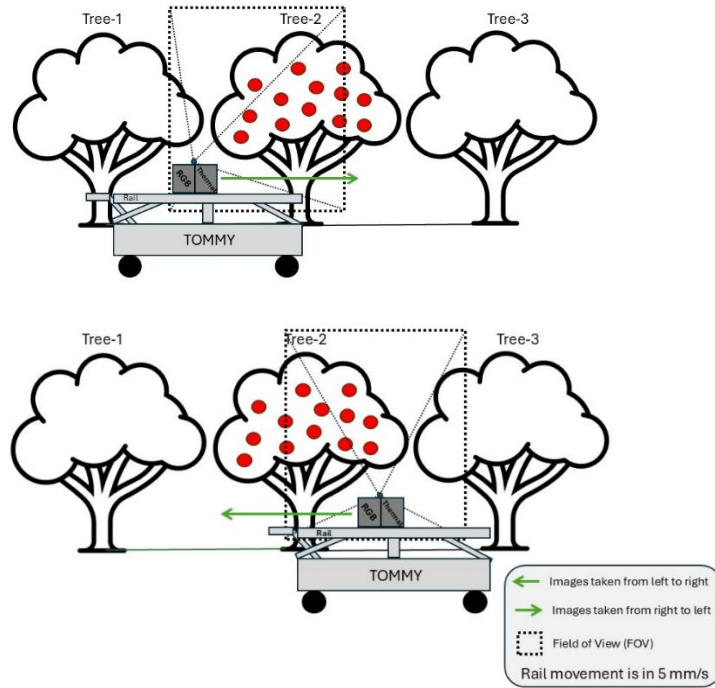
Imaging was performed under three conditions: at night with a manual light source, at sunrise with dew on the fruits and during daylight when fruits were dry. On average, about 4400 in 2024 and 2500 in 2025 acquisitions were captured per session, conducted twice a week, approximately every two weeks during the acquisition period. Data were collected across five acquisition time points in 2024 and three acquisition time points in 2025. In both 2024 and 2025, TOMMY was positioned along the field to capture images ensuring full canopy coverage (Fig. 3). For each tree imaged from one side of the field, corresponding images were also collected from a tree on the opposite side in consecutive days, ensuring complete spatial representation.

Table 1

Overall data collection 2024 and 2025 parameters.

Year	2024	2025
Number of trees per session	12	
Number of Images Per Tree Each Session	40-60	
Trees Selected	Trees with -30% and +20% irrigation	
Number Of Orchards	1	
Growing Conditions	Excessive and non-excessive watering external humidity, dry temperatures	
Rounds	8	6
Total RGB Images (Data)	16,922	7481

Several disruptions in 2024 prevented a consistent timeline. To maintain uniform three-time points structure across trees, two artificial time points were created, and one of the time points was removed due to missing trees. After filtering incomplete cases, 9,276 RGB images remained, forming 3,092 complete three-time-point sequences (maximum of six images per sample: [north + south] $\times$ 3), which were used as inputs for predicting the fourth time point. The same preprocessing was applied in 2025. After alignment and removing incomplete trees, 3,024 RGB images organized into 1,008 three-time-point sequences were created and used as an independent test set.

**Fig. 3.** Schematic diagram of data acquisition in the field.**Table 2**

Specifications of the thermal and RGB sensors used for fruit monitoring.

Sensor	Thermal	RGB
Name	Flir A655sc	Canon EOS R5
Lens	35 mm	35 mm
Resolution	640X480	8192X5464
Sensitivity	<30 mK NETD	ISO 100-51,200 (dynamic range ~14 stops)

Accuracy	$\pm 2\text{ }^{\circ}\text{C}$ or $\pm 2\%$ of reading	Not specified (calibration-dependent)
F-number	f/1.0 (35 mm lens)	Depends on lens (e.g., f/1.8 for RF 35 mm)
Spectral Range	7.5-14 μm	Visible light

2.1.1. Image acquisition system

TOMMY is a ground-based measurement system designed for field data collection and analysis (Fig. 4). The system integrates multiple sensors, including a thermal camera (FLIR A655sc) and an RGB camera (Canon EOS R5), to capture detailed visual and thermal information (camera specifications are provided in Table 2). The cameras are mounted on a motorized linear conveyor that moves along the x-axis at a constant speed of 5 mm/sec. During acquisition, images are captured automatically every 5 seconds, resulting in a spatial separation of approximately 2.5 cm between consecutive images along the canopy. This acquisition scheme provides dense and overlapping coverage of each tree, ensuring consistent spatial sampling across trees and sessions. System operation, calibration, and data recording are controlled through a dedicated interface (ATB software), enabling synchronized image acquisition and reliable field operation. Examples of RGB and thermal images collected from Wonderful pomegranate cultivar trees using the TOMMY system during the 2024 season are shown in Fig. 5.



Fig. 4. Photographs of the TOMMY system (a) Cameras mounted on the TOMMY vehicle (b) TOMMY vehicle.

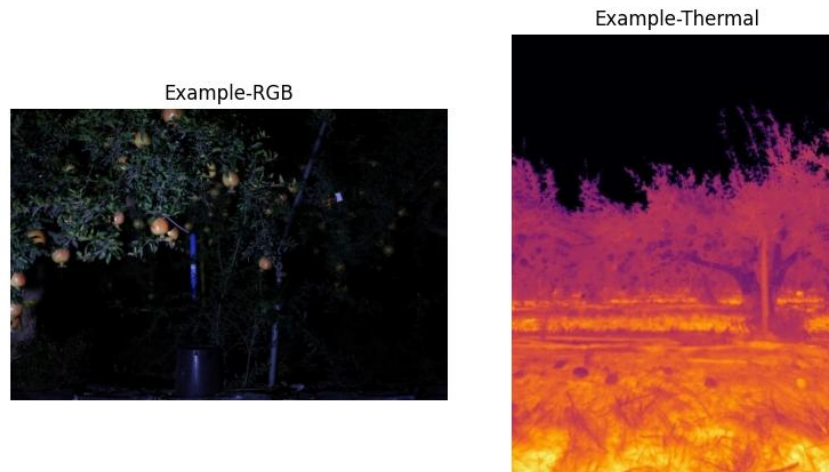


Fig. 5. Example RGB nighttime image (a) and thermal image (b) of Wonderful pomegranate cultivar trees captured using the TOMMY system in 2024.

2.1.2. Meteorological station data

Because weather significantly influences agricultural variability (Hoogenboom, 2000), environmental data were collected using a meteorological station that recorded measurements every

minute during data acquisition. The recorded variables included the calendar date, time of measurement (hh:mm), average air temperature ($^{\circ}\text{C}$), average relative humidity (%), average solar radiation ($\text{W}\cdot\text{m}^{-2}$), average canopy temperature ($^{\circ}\text{C}$), and average dew point temperature ($^{\circ}\text{C}$).

2.1.3. Extraction of tabular thermal data

Image feature extraction has been emphasized in decades of research as a key determinant of registration accuracy (Keyvanpour et al., 2021). Because image-to-image fusion calibration between the RGB and thermal cameras in 2024 was imperfect, pixel-level alignment was not possible. This was mainly due to differences in focal length and field of view between the sensors, with the thermal camera having a wider lens than the RGB camera lens. As a result, thermal images were converted to TIFF, cropped into the fruit region, and processed into tabular features using statistical, gradient-based, entropy, and frequency-domain descriptors, with the same procedure applied in 2025 for consistency. The extracted features included mean temperature ($^{\circ}\text{C}$), representing the average thermal value across all pixels, skewness, describing the asymmetry of the temperature distribution, gradient mean, capturing the average magnitude of local temperature changes between neighboring pixels, Haralick energy, reflecting textural uniformity within the thermal image, and image entropy, representing the overall randomness or complexity of pixel intensity values.

2.2. Ground truth and heuristic masks labeling for preprocessing

An external dataset published by (J. Wang et al., 2024), containing 13,840 annotated pomegranate bounding boxes (PG-YOLO, GitHub), was used to initialize the object detection model through transfer learning, reducing the need for extensive manual annotation. To adapt the model to our orchard, 616 images from our dataset (Average of 60 fruits per image) were manually labeled using CVAT and used for fine-tuning. After object detection model training the resulting bounding boxes were then passed to the Grab-Cut algorithm to generate heuristic segmentation masks. Grab-Cut models the image (RGB image) foreground and background the with Gaussian Mixture Models (GMMs) (Rother et al., 2004) to refine object boundaries and create binary masks, due to the high resolution and iterative nature of the method, segmentation was performed on 869 representative images. Because no ground-truth segmentation labels were available, mask quality was assessed using the Structural Similarity Index Measure (SSIM), which evaluates similarity in luminance, contrast, and structural patterns between two images (Z. Wang et al., 2004). SSIM values range from 0 to 1, with higher scores indicating greater structural fidelity. The mean SSIM across all image-mask pairs was used to quantify segmentation quality, and the resulting scores are reported in (Peleg., A., 2026).

2.3. Tree pomegranate cracking prediction labeling schemes

Tree pomegranate fruit cracking prediction was formulated using three labeling schemes summarized in [Table 3](#): multi-class tree cracking severity, multi-class fuzzy tree cracking severity, and tree binary cracking labels. For the severity-based schemes, trees were labeled according to the number of cracked fruits observed at the final available time point, with class 0 representing 0-2 cracked fruits, class 1 representing 3-5 cracked fruits, and class 2 representing more than 6 cracked fruits. Fuzzy severity labels were derived from the same class intervals by assigning soft membership values based on proximity to predefined class medians, defined as 1, 4, and 8 cracked fruits for classes 0, 1, and 2, respectively, allowing partial membership near class boundaries. In addition, a binary labeling scheme was defined, where trees were labeled as 0 if no cracking was observed and 1 if any degree

of cracking occurred. Manual labeling was performed by the researcher for a total of 3,092 sequences in 2024 dataset and a total of 1008 sequences in 2025 data set, each composed of three consecutive time points, using the final time point of each sequence as the labeling reference ([section 2.1](#)). The goal is to predict the cracking severity of each tree based on data from previous time points.

Table 3
Summarized cracking prediction labeling schemes.

Property	Multi-class tree cracking severity	Fuzzy tree cracking severity	Tree binary cracking
Predicts cracking severity level	V	V	X
Predicts cracking occurrence	V	V	V
Soft probabilistic labels	X	V	X
Hard discrete class labels	V	X	V
Number of output classes	3(hard)	3 (soft)	2(hard)
Intended use case	Cracking severity estimation	Fuzzy cracking severity estimation (Uncertainty-aware)	Early Cracking prediction

2.4. Model performance evaluation measures

The Following metrics were used to evaluate the performance of the different models: Recall [Eq.\(1\)](#) Precision [Eq.\(2\)](#) and F1-Score [Eq.\(3\)](#). The main objective is to maximize all three measures; however, given the trade-off between precision and recall, priority is placed on recall to ensure the correct label is identified. Additionally, each DL model had a main evaluation metric to evaluate accuracy of predictions, which is explained separately in the subsections below.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

$$\text{F1score} = \frac{2TP}{2TP+FP+FN} \quad (3)$$

where TP is True Positive, TN is True Negative, FP is False Positive, and FN is False Negative.

2.4.1. Object detection model evaluation

Mean Average Precision (mAP) [Eq.\(4\)](#) was used to evaluate the object detection model. Performance was reported using mAP50, computed at an IoU threshold of 0.50, and mAP50-95, averaged over Intersection over Union (IoU) thresholds from 0.50 to 0.95 in steps of 0.05, providing a stricter assessment of localization accuracy.

$$\text{mAP} = \left(\frac{1}{N} \sum_{i=1}^N AP_i \right) \quad (4)$$

where N is the number of object classes and AP_i is the Average Precision for class i .

2.4.2. Segmentation model evaluation

For segmentation evaluation, IoU [Eq.\(5\)](#) was used as the primary metric. IoU measures the overlap between the predicted and ground truth.

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (5)$$

where Area of Overlap is the intersection between the predicted and ground-truth masks, and Area of Union is the total area covered by both masks.

2.4.3. CNN+LSTM for prediction of cracking severity evaluation

The model in both cracking severity labels (section 2.3.1) and binary labeling (section 2.3.3) was evaluated using accuracy Eq.(6) metric of the last time point (i.e., the predicted value on the last time point). This approach assessed how well the model captures temporal patterns and ensures consistent performance in predicting cracking overtime. Additionally, an Agreement Accuracy (AA) (Peleg., A., 2026) Eq.(7) metric was created to evaluate consistency between the two tree sides (Fig. 2), meaning that predictions from data at the north side of the field and south side of the field (section 2.1) for the same tree must be similar and assigned the correct label. This metric represents a stricter form of accuracy, as it requires agreement across both sides of the tree to be considered correct.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN} \quad (6)$$

$$AA = \frac{\text{Agreement between north and south same samples}}{\text{All samples}} \quad (7)$$

2.4.4. Fuzzy labeling evaluation metrics

To evaluate the performance and consistency of the fuzzy label (section 2.3.2) predictions, three complementary metrics were used. Mean Squared Error (MSE) Eq.(8) quantifies how far the predicted fuzzy vector is from the expected soft label. Inter-side MSE metric (Peleg., A., 2026) Eq.(9) was used to compare predictions from the north and south sides of the field for the same tree, ensuring that both viewpoints produce similar outputs, as they observe the same fruit from different angles. Finally, to enable a numeric comparison between soft (fuzzy) labels and hard labeling schemes (multiclass and binary), a composite accuracy (CA) score was defined by combining fuzzy-label agreement, computed as one minus the normalized MSE Eq.(10) with hard-label classification of fuzzy-label accuracy Eq.(7) using a weighted-sum formulation Eq.(11) with equal weighting ($\alpha = 0.5$) (Peleg., A., 2026). Together, these metrics assess fuzzy prediction accuracy, cross-view consistency, and agreement with hard-label performance.

$$MSE = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{C} \sum_{i=1}^C (p_i - t_i)^2 \right] \quad (8)$$

where p_i is the predicted probability for class i , t_i is the target fuzzy membership for class i , N is the number of samples and C is the total number of classes.

$$MSE(\vec{p}_{\text{North}}, \vec{p}_{\text{South}}) \quad (9)$$

where $\vec{p}_{\text{north}}$ and $\vec{p}_{\text{south}}$ are the fuzzy prediction vectors from the north and south sides of the field for the same tree.

$$MSE_{\text{norm}} = \frac{MSE}{2/C} \quad (10)$$

where C is the number of classes, and $2/C$ represents the maximum possible MSE Eq.(8) obtained when the predicted and target vectors are one-hot vectors assigned to different classes.

$$CA = \alpha \cdot (1 - MSE_{\text{norm}}) + (1 - \alpha) \cdot \text{Accuracy} \quad (11)$$

where $\alpha \in [0,1]$ controls the relative contribution of fuzzy-label agreement and hard-label classification accuracy.

3. Algorithm

This chapter details the DL algorithms applied throughout the preprocessing framework and model combined with data analysis for feature engineering.

3.1. Algorithm framework

The proposed multimodal framework ([Fig. 6](#)) represents the final pipeline developed for predicting pomegranate tree fruit cracking severity using RGB imagery, thermal data, and meteorological measurements. An object detection model based on YOLOv8 was first trained using 616 manually annotated RGB images together with external PG-YOLO data and then fine-tuned using transfer learning (TL) to improve detection accuracy ([section 3.2](#)). Cropping was applied to the RGB and thermal images to focus on the fruit region and reduce background noise from sky and ground. The fine-tuned YOLOv8 model was then used to generate bounding boxes from RGB images, which were used to create 869 Grab-Cut algorithm masks ([section 2.2](#)) (Rother et al., 2004) for the segmentation stage. A U-Net-based segmentation model (Ronneberger et al., 2015) was trained to isolate pomegranate regions from resized RGB images, with data augmentation applied to improve generalization by expanding the 869 cropped samples to 1,849 through horizontal flipping, scaling, translation, rotation, and random cropping (Khalifa et al., 2022). In parallel, thermal images were converted to tabular features ([section 2.1.3](#)), and meteorological data were collected from a field station ([section 2.1.2](#)). Both data sources were internally aligned using timestamp matching and standardized using z-score normalization. After segmentation, each RGB image was matched with its corresponding thermal and meteorological records to form multimodal inputs, followed by feature engineering and dimensionality reduction after multicollinearity checks that was done in (Peleg., A., 2026). Prior to prediction, segmented RGB images are enhanced through image processing techniques ([Fig. 9](#)), including grayscale conversion, CLAHE enhancement (Pizer et al., 1990; Reza, 2004), Gaussian blurring (Cabello et al., 2015; Gedraite & Hadad, 2011), and Laplacian edge detection (Kaur et al., 2008; Mutneja, 2015), to enhance spatial texture representation. To predict cracking severity before it occurs a CNN+LSTM architecture is employed, the CNN extracts spatial features from the processed RGB images and the LSTM captures temporal patterns in the aligned thermal and meteorological data. Model training and optimization were conducted exclusively on the 2024 dataset under different labeling schemes ([section 2.3](#)) (Peleg., A., 2026). Data from 2025 were used only in the generalization phase to evaluate cross-season performance across all labeling schemas ([section 2.3](#)), based on the optimized CNN+LSTM model that yielded the best results in terms of accuracy ([section 2.4.3](#)). For this evaluation, preprocessing of the 2025 RGB images relied entirely on the segmentation model trained in 2024, and tabular data alignment followed the same procedure as in the 2024 framework. All model development, training, preprocessing, data manipulation, statistical tests, and experimentation were performed in the Google Colab Pro environment using an NVIDIA A100 GPU.

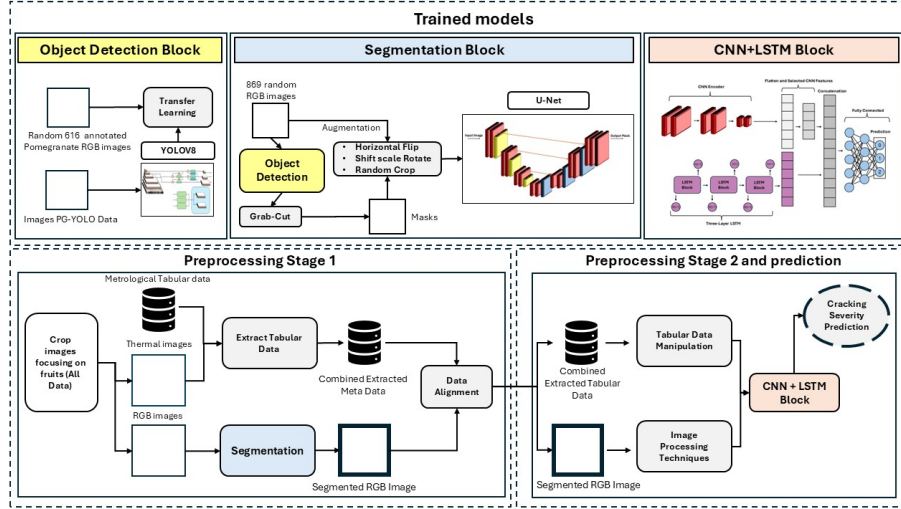


Fig. 6. Framework of algorithms until cracking severity prediction.

3.2. YOLOv8 object detection model

The YOLO architecture (Redmon et al., 2015) (Fig. 6) was used for pomegranate detection, with the YOLOv8-s variant (11.1M parameters) selected for its balance of accuracy and efficiency (Ultralytics). Although YOLOv8 is pretrained on COCO (Lin et al., 2014), no COCO data were used, all training relied solely on pomegranate images. YOLOv8 processes images through a multi-scale backbone, a neck that combines and up-samples features, and a head that predicts bounding boxes and class probabilities. Model development followed a two-stage approach: first, YOLOv8 was trained from scratch on the PG-YOLO dataset (J. Wang et al., 2024), containing 13,840 annotated images, using 50 epochs (batch size 16) with 9,798 training and 4,042 testing images. This provided general pomegranate-specific features. Second, 616 images from our dataset (section 2.2) were manually annotated and used for fine-tuning to adapt the model to orchard lighting, background, and viewpoint variability. Raw transfer evaluation of the PG-YOLO model on our dataset showed low performance (section 4.1), confirming the need for inductive transfer learning. For fine-tuning, 500 images were used for training and 116 for testing over 50 epochs (converging at epoch 38). Experiments with different train-test splits showed that 80/20 yielded the best performance and was therefore selected. This two-step procedure enabled an accurate and computationally efficient detection model tailored to orchard conditions.

3.3. U-Net Segmentation Model

The U-Net architecture (Ronneberger et al., 2015) (Fig. 6) was used for the segmentation task due to its encoder-decoder structure and skip connections, which make it effective with limited data. In this architecture (Fig. 13), the encoder extracts features using convolutional layers, ReLU activations, batch normalization, and max pooling, while the decoder up-samples feature maps with transposed convolutions and refines them through skip connections to recover fine spatial details. The model outputs raw logits, which are later thresholded to produce binary masks of pomegranates. U-Net was implemented with an EfficientNet-B7 backbone (Tan & Le, 2019), leveraging compound scaling for improved accuracy and computational efficiency. Images were resized to 160×256 to meet U-Net's 32-pixel down-sampling constraints. Training used CrossEntropyLoss with Grab-Cut masks (section 2.3.5), AdamW optimization (learning rate 0.001, weight decay $1e-4$), and a ReduceLROnPlateau scheduler (factor 0.5, patience 2) and the model was trained for 30 epochs. The segmentation dataset contained 869 images with corresponding masks, and augmentation expanded it

to 1,849 image mask pairs used for training and testing. Both training from scratch and initializing the encoder with ImageNet-1K pretrained weights (Deng et al., 2009) were evaluated, based on prior findings that pretrained CNN features often improve performance on limited datasets (Igllovikov & Shvets, 2018; Yosinski et al., 2014). Additional experiments evaluated U-Net configurations using ResNet-50 and EfficientNet-B7 backbones, with and without augmentation. The EfficientNet-B7 model with ImageNet-pretrained weights and augmentation achieved the best performance and was selected for this study, full experimental results appear in (Peleg., A., 2026).

3.4. Final CNN+LSTM cracking prediction model

CNN+LSTM is a hybrid DL model that combines convolutional neural networks (CNNs) for spatial feature extraction from images with long short-term memory block (LSTM) networks block for capturing temporal dependencies across sequential data (Fig. 7).

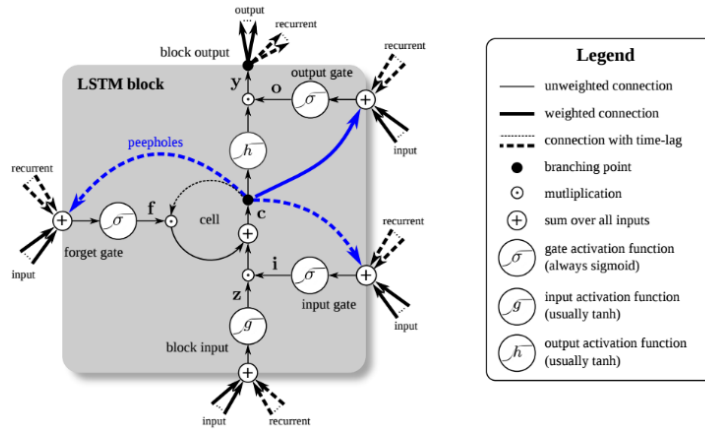


Fig. 7. Detailed schematic of LSTM as used in the hidden layers of a recurrent neural network (Greff et al., 2015).

Intermediate (feature-level) multimodal fusion strategy (Fig. 8) was selected, allowing each modality to be encoded independently before being combined in a fully connected prediction layer. Following the formal definition of intermediate fusion described by (Guarrasi et al., 2025), each modality x_i is first processed by its modality-specific function f_i to obtain an intermediate representation $h_i = f_i(x_i)$. The representations are then combined using a fusion function F Eq.(12).

$$\mathbf{h} = \mathcal{F}(\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n) \quad (12)$$

The fused representation $\mathbf{h}$ is subsequently passed through a final function f to generate the output prediction Eq.(13).

$$\mathbf{y} = \mathbf{f}(\mathbf{h}) \quad (13)$$

where x_i denotes the input of modality i (such as RGB images or tabular thermal-meteorological data), f_i represents the modality-specific feature extractor applied to that input (a CNN for images and an LSTM for sequential tabular data), and $h_i = f_i(x_i)$ corresponds to the intermediate feature representation produced for each modality. The fusion function F combines these intermediate representations implemented here as simple concatenation to produce the joint feature vector $\mathbf{h}$, which is then passed to the final decision function f to generate the output prediction $\mathbf{y}$.

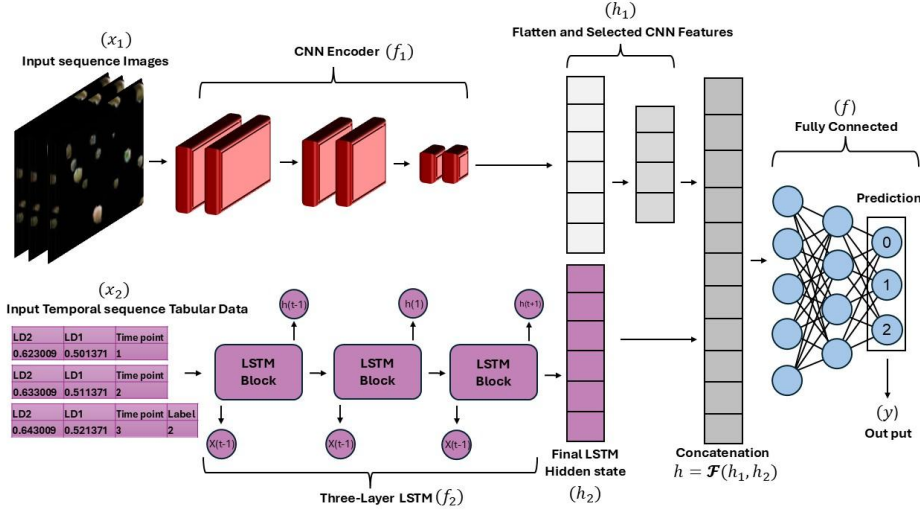


Fig. 8. CNN + LSTM general intermediate fusion architecture.

3.4.1. CNN+LSTM model overview

The model (Table 4) integrates ConvNeXt backbone for image feature extraction and incorporates two attention mechanisms: a spatial attention block in the CNN branch and a temporal attention block in the LSTM branch. Feature engineering includes feature creation, with linear discriminant Analysis (LDA) applied to the combined thermal and environmental tabular data to generate new discriminative features that replace the original ones. The image-processing pipeline consists of grayscale conversion with CLAHE, followed by Gaussian blurring and Laplacian edge detection to enhance fruit-surface texture patterns.

Table 4
Summarized model overview.

Category	Component
CNN Backbone	ConvNeXt
Feature Engineering	Feature Creation: LDA on tabular combined data were features were selected and created in before models
Attention	Sperate CNN and LSTM attention blocks
Image Processing	Gray scale transformation + CLAHE equalization + Gaussian blurring + laplacian edge detection
Model Architecture	CNN+LSTM intermediate fusion architecture
Loss Function	Focal loss

In the proposed intermediate fusion CNN+LSTM architecture, two streams process different modalities: RGB images (image input size of 224x224) and environmental and thermal tabular data. The CNN extracts spatial features from each RGB image across three time points. These features, originally in the form of multi-dimensional tensors, are flattened meaning they are reshaped into one-dimensional vectors to prepare them for concatenation with the tabular data-based features. Flattening is necessary because fully connected (FC) layers, which follow, require fixed-length vector inputs rather than multidimensional image feature maps. In parallel, the LSTM processes tabular data sequences corresponding to the same time points. The LSTM produces a final hidden

state vector (not a sequence), which is the output of the last LSTM cell summarizing information from all previous time points. This vector already has the proper flattened shape and captures the temporal dependencies within the structured tabular data. After flattening the CNN outputs and obtaining the LSTM's final hidden state, the two representations are concatenated to form a single feature vector that merges spatial and temporal information. This combined vector is passed through a series of FC layers that are responsible for serving as the final stage that transforms the combined image and tabular data features into class predictions. It contains in our case Linear layers that reduce the dimension of the input, nonlinear activation layers, and dropout layers for regularization are applied. Dropout randomly deactivates a portion of the neuron outputs during training to prevent overfitting and encourage the model to learn more robust and generalized patterns. These FC layers reduce the dimensionality of the combined feature vector and help the model learn joint representations by capturing complex relationships between the image and tabular data inputs.

The fully connected (FC) layer structure in our model remained the same across all configurations. The input to the FC block is a concatenated feature vector of size (CNN outputs $\times$ 3 for the three image time steps + LSTM output), representing the flattened CNN outputs together with the LSTM's final hidden state. This vector first passes through a Linear layer that reduces its dimensionality to 512, followed by a ReLU activation and a dropout layer with a rate of 0.3. The next stage repeats this pattern with a Linear layer reducing the dimension from 512 to 256, followed by another ReLU activation and a dropout rate of 0.2. The final Linear layer maps the 256 features to the number of classes, producing raw logits that represent the model's prediction scores before normalization. These raw logits are then passed to a soft-max function Eq.(14) to compute class probabilities. The soft-max function transforms these raw logits into a probability distribution over the classes by exponentiating each score and normalizing by the sum of all exponentiated scores, ensuring the output values lie between 0 and 1 and sum to 1. These predictions correspond to the model's estimation of fruit cracking severity. SoftMax function is defined as:

$$p_i = \frac{\exp(\alpha s_i)}{\sum_j \exp(\alpha s_j)} \quad (14)$$

where α is a parameter that modulates the shape of the output distribution by systematically influencing how the logits are scaled before normalization, s_i are the input scores produced by the model (also referred to as logits) and p_i are the resulting normalized probabilities after applying the soft-max function.

3.4.2. Dataset and training configuration

The dataset split described here (Table 5) was used on the 2024 data, while generalization process on the 2025 season is explained separately in section 3.4.3. The dataset was split into training, validation, and test sets following: 80% of the data was used for training, and 20% (randomly chosen) was reserved for testing. From the training data, an additional 20% was set aside as a validation set to monitor model performance during training. For this model the north side of the field and the south side of the field data were combined in to one dataset for training and testing to allow the model to learn from a larger and more diverse dataset, resulting in 1960 training samples, 490 validation samples, and 642 test samples. During training, the model was run for 30 epochs and the weights achieving the lowest validation loss were saved and used for final testing to ensure the evaluation reflected the best-performing configuration. Additionally, sensitivity analyses were

performed to evaluate the robustness of the CNN+LSTM model, including systematic variation of batch size, learning rate, and the image-processing pipeline (CLAHE clip-limit, Gaussian blur kernel size, and Laplacian edge strength). These analyses guided the selection of the optimized model configuration presented in this study, and full results are provided in (Peleg., A., 2026). Additionally, in (Peleg., A., 2026), statistical tests were conducted prior to selecting the optimized model to verify that prediction performance from the north and south sides of the field was not significantly different, thereby allowing both sides to be merged into a single unified dataset. Using the same model architecture both fuzzy labeling ([section 2.3.2](#) and [section 3.5.1](#)) training with MSE loss function ([section 3.5.1](#)) and binary labeling ([section 2.3.3](#) and [section 3.5.2](#)) were applied to test how the model performed under different ways of defining class boundaries, as these approaches more accurately reflect the uncertainty inherent in the problem and help ensure that labels correspond more closely to the actual characteristics of each case. Fuzzy labeling was used because the original class intervals were created manually for this research, without any established reference, meaning the limits between categories were invented and could be different in real-world applications. To reflect this uncertainty and the possibility that class definitions might shift, continuous values between 0 and 1 were assigned to each sample based on the distance between the actual number of cracked fruits on the tree and the median cracking severity of each class interval. In that scenario class label distribution and train test split stayed the same. In contrast, binary labeling simplified the classes into only two real world scenario categories, such as cracked fruits on tree (1) and not cracked (0), to test how well the model could make a clear decision. This binary approach could also be useful as a practical early prediction step to flag trees that will have fruits that will crack.

Table 5

Summarized 2024 dataset and training configurations for all labeling schemes.

Field Side	Train Samples	Validation Samples	Test Samples	Batch Size	Epochs	Learning Rate
North + South (combined)	1960	490	642	8, 16, 32, 40	30	0.001, 0.00055, 0.0001

3.4.3. 2025 dataset generalization test

Model generalization across seasons was evaluated for each labeling scheme ([section 2.3](#)) by examining performance degradation, that is, how much the metrics ([section 2.4](#)) increased or decreased when applied to the 2025 dataset. All optimized labeling schemes were first trained and evaluated on the 2024 dataset, and these same optimized models were then applied directly to the independent 2025 dataset for generalization testing. Since the 2024 dataset contains 3,092 three-time-point sequences and the 2025 dataset contains 1,008 sequences, the 2024 data were used as the training/validation source with same validation percent split as before ([section 3.4.2](#)) and the 2025 data were used as the external test set. Generalization was therefore evaluated only in the forward direction (train/validation 2024, test 2025).

3.4.4. CNN+LSTM model

Two separate attention blocks were implemented in the model architecture to enhance temporal feature aggregation: one for the LSTM-based tabular data sequence (1) designed to operate on the

output of an LSTM sequence, this attention block receives a tensor of shape (B,T,H) where B is the batch size, T is the number of time steps, and H is the LSTM's hidden size. Each time step's output vector is passed through a fully connected linear layer applied independently to produce a scalar attention score per time step. This scoring layer learns to capture the relative importance of each point in the sequence. The resulting scores retain the shape (B,T,1) (1 indicates that each time step is assigned a single attention score), preparing them for the soft-max normalization and weighted sum used to form a context vector. and another for the CNN-based image feature sequence (2) designed to operate on a sequence of image feature vectors extracted from a convolutional neural network (CNN), where each vector corresponds to a different time point. The input is a three-dimensional tensor of shape (B,T,D), where B is the batch size, T is the number of time steps (e.g., time points), and D is the dimensionality of the CNN features. To compute the attention scores, each time step's feature vector is passed through a small multilayer perceptron (MLP), a feedforward neural network consisting of two fully connected layers with a non-linear RELU activation function in between. The MLP transforms each feature vector into a single scalar score that reflects its relevance within the temporal sequence. This allows the network to capture non-linear dependencies between image features across time. The resulting scores retain a shape of (B,T,1) (1 indicates that each time step is assigned a single attention score) and are then used to generate attention weights and a context vector through a weighted sum. The attention operations described below are based on the additive attention mechanism proposed by (Bahdanau et al., 2014) originally developed for word alignment in machine translation. In our case, this formulation is adapted to emphasize temporal relevance by assigning attention weights across discrete time points. Soft-max normalization Eq.(15) Converts raw attention scores into a probability distribution over time steps, highlighting the most relevant inputs and context vector calculation Eq.(16) Computes a weighted sum over the sequence of inputs using attention weights, focusing on the most informative parts of the input.

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)} \quad (15)$$

where α_t is the attention weight at time step t, e_t is the raw attention score at time step t, e_k represents the raw attention score for time step k (referring to all scores used in the soft-max normalization) and T is the total number of time steps.

$$c = \sum_{t=1}^T \alpha_t \cdot x_t \quad (16)$$

where c is the output context vector, α_t is the attention weight from the soft-max, and x_t is the feature vector at time step t.

For better class separability LDA was used. To do this, Dunn's post-hoc test was used. Dunn's test is in (Peleg., A., 2026) a non-parametric pairwise comparison method used after detecting overall group differences and helps identify exactly which pairs of groups differ significantly. From Dunn's test, it was clear that for each feature, at least one time point did not show significant differences between the class labels. This lack of class separability can reduce model learnability, as features that fail to distinguish between classes introduce noise and reduce classification effectiveness. To address this, LDA was applied as a supervised dimensionality reduction method (section 4.4.2). (Tharwat et al., 2017) Explains it like this: LDA aims to project high-dimensional data onto a lower-dimensional space that best separates the classes. The first step involves calculating the between-class variance, which measures the spread between the means of different classes. The second step computes the within-class variance, capturing how much the individual samples vary around their class mean. In

the third step, LDA finds the projection that maximizes the ratio of between-class to within-class variance, ensuring that the classes are as distinct as possible in the reduced space. This is known as Fisher’s criterion, and it aims to maximize the separability between classes while minimizing the spread within each class. This optimization leads to the generalized eigenvalue problem. The full results of this process are shown in (Peleg., A., 2026).

The ConvNeXt-Base configuration was selected as the backbone and used with ImageNet-1k pretrained weights as the image encoder (Dosovitskiy et al., 2020; Liu et al., 2022). The architecture of the model is structured like this: Input images arrive in a 5D tensor $(B, 3, C, H, W)$ like before and are reshaped into $(B \times 3, C, H, W)$ before being processed by the ConvNeXt backbone. Unlike ResNet, where CNN features from all three-time steps were flattened into a single long vector $(B, 3 \times 2048)$, the ConvNeXt outputs are kept in temporal order and reshaped to $(B, 3, F)$ where F is the feature dimension (e.g., 1024), allowing attention to operate across time. The image attention block then reduces this to a context vector of shape (B, F) , summarizing the most relevant time step. On the tabular data side, input with shape $(B, 3, D)$ is passed through 3 LSTM blocks, and instead of using only the final hidden state, the hidden output at all time steps $(B, 3, \text{hidden dim})$ is retained and processed through a tabular data attention block to generate a weighted context vector $(B, \text{hidden dim})$. These two vectors are concatenated into a single vector $(B, F + \text{hidden dim})$ and passed through FC classification head used in the original model.

Training begins by computing normalized class weights from the training data to address class imbalance. The training labels are collected, the frequency of each class is counted, and a weight is calculated for each class as the total number of samples divided by the product of the number of classes and the number of samples in that class. This ensures that classes with fewer examples receive higher weights, allowing the loss function to give underrepresented classes more influence during training. The weights are then normalized so their average is 1, ensuring balanced contribution to the overall loss without disrupting the optimization process. These weights are used in the Focal Loss function, which modifies standard cross-entropy to focus training on hard-to-classify examples. Focal Loss introduces two key components: a focusing parameter ($\gamma = 2.0$), which increases emphasis on misclassified samples by down-weighting well-classified ones, and a scaling factor ($\alpha = 1.0$), which globally balances class importance in imbalanced datasets. During the forward pass, the method computes the cross-entropy loss for each sample without reduction, applies soft-max to obtain class probabilities, and uses one-hot encoding to isolate the correct-class probability p_t . A modulating factor $(1 - p_t)^\gamma$ reduces the loss contribution of confident predictions so that misclassified or ambiguous samples receive more focus. This value is then scaled by α and multiplied by the original cross-entropy to compute a focal-weighted loss per sample, which is averaged across the batch. Class weights derived from label frequencies are included to further address imbalance. This approach follows the formulation introduced by (Lin et al., 2017). The Adam optimizer (very similar to AdamW from [section 3.3](#), but without applying weight decay separately, so it doesn't penalize large weights as effectively) is used for parameter updates with learning rate of 0.001, and ReduceLROnPlateau ([section 3.3](#)) dynamically adjusts the learning rate by monitoring the validation loss, reducing it in factor of 0.5 if no improvement is seen for two epochs. As part of the model’s data preprocessing stage, the RGB images underwent a unified image-processing pipeline designed to enhance fine-grained fruit-surface textures before feature extraction. Each image was first converted to grayscale to emphasize structural details, then

enhanced with CLAHE to improve local contrast. To further strengthen texture representation, Gaussian blurring was applied to reduce noise, followed by Laplacian edge detection to highlight subtle surface boundaries related to cracking ([Fig. 9](#)). All techniques were applied together as a single preprocessing framework.

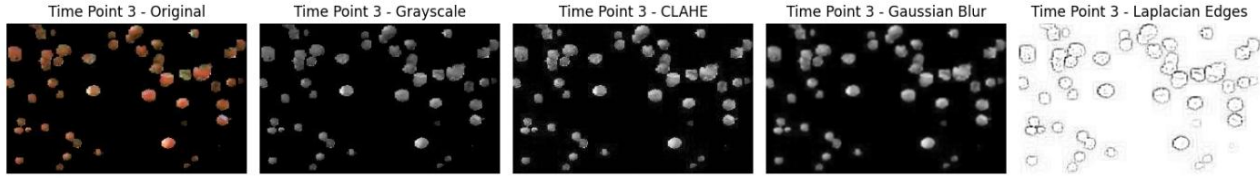


Fig. 9. Applied image processing on one of the times point sequences: (a) original RGB image, (b) grayscale transformation, (c) CLAHE equalization (d) gaussian blur and (e) laplacian edges.

3.5. Label creation tests

3.5.1. Fuzzy label creation

To better represent uncertainty and gradual changes between cracking severity levels, fuzzy labels ([section 2.3.1](#)) were created based on the number of cracked fruits per tree considering the median of each class interval. Unlike hard labels, which assign a single class to each sample ([section 2.3.2](#)), fuzzy labels give each sample a soft probability vector showing how strongly it belongs to each class. This allows for partial membership and reflects the fact that some cases are not clearly one class or another. Gaussian distance fuzzy membership was used [Eq.\(17\)](#) it gives each class a smooth bell-shaped score curve, where nearby classes get higher values and distant ones get smaller values. The spread of the curve is controlled by $\sigma = 2$ in our case making the transition between classes soft and gradual.

$$\text{Gaussian Distance} = \exp\left(-\frac{(x-c_k)^2}{2\sigma^2}\right) \quad (17)$$

where x , c_k , and s_k are the same as defined previously (Inverse distance), and σ is the standard deviation that controls how sharp or soft the curves are.

After creating the labels normalization was done [Eq.\(18\)](#) This converts raw similarity values into fuzzy label vectors, where each value represents the proportion (or "probability-like" score) for that class.

$$y_k = \frac{s_k}{\sum s} \quad (18)$$

Where $\sum s$ is the sum of all scores across classes.

Additionally in the fuzzy labeling setup, classification-based loss functions such as Focal Loss ([section 3.4.4](#)) were not suitable, as they require discrete class labels and operate on categorical probability distributions. Fuzzy labels, however, are continuous values between 0 and 1, designed to reflect uncertainty around class boundaries. This makes the task more similar to a regression problem, as the model predicts continuous values rather than discrete classes. Accordingly, MSE loss was used to encourage accurate numerical predictions. MSE computes the average of the squared differences between predicted outputs and target values, penalizing larger errors more heavily and encouraging the model to produce numerically accurate predictions that align closely with the soft labels.

3.5.2. Binary label creation

As mentioned before ([section 3.4.1](#)), in this setup the target is to simplify the classes into only two real-world scenario categories: cracked fruits on the tree (1) and not cracked (0). This binary division reflects the reality observed in the field, where trees typically showed cracking symptoms or remained unaffected. In the binary classification setting ([section 2.3.3](#)), the focal loss ([section 3.4.4](#)) was adjusted to align with the two-logit soft-max output structure. Since the model outputs class probabilities over two classes using soft-max, the ground truth labels (0 or 1) were converted to one-hot encoded vectors to match the prediction dimensions. This allowed for extracting the true class probability by computing the element-wise product between the one-hot encoded targets and the soft-max probabilities, followed by a summation over the class dimension. It's important to note that the binary labels were created after the original three-class setup, the LDA step was not repeated for the binary case. Instead, the LDA features generated from the three-class labels were used directly. Recomputing LDA with binary labels would yield only one feature, so keeping the multi-feature LDA may slightly influence results but provides a richer representation.

4. Results and discussion

4.1. Object detection YOLOv8 model results

All YOLOv8 object detection experiments are described in [section 3.2](#), and the results are summarized in [Table 6](#). The PG-YOLOv8 data model achieved strong detection performance, with an mAP50 of 93.8%, mAP50-95 of 59.7%, precision of 88.1%, recall of 86.8%, and an F1-score of 87.0%, indicating effective detection and localization of pomegranates. In contrast, the raw transfer experiment, in which the PG-YOLOv8 model trained on the PG-YOLO dataset was applied to our dataset without fine-tuning, resulted in a substantial performance decline, with mAP50 and mAP50-95 dropping by 47.2% and 22.6%, respectively (-46.6% and -37.1%). While precision remained relatively high at 75.4%, recall decreased sharply (-53.7%), indicating that many pomegranates were not detected. In the left side of [Fig. 10](#), many clearly visible pomegranates are not detected, highlighting the limitations of the pre-trained model when applied to our specific dataset. This issue arises from significant differences in data distributions between datasets, such as variations in lighting conditions, tree structure, fruit placement, and environmental conditions. These discrepancies emphasize the need for TL to adapt the model to our dataset and improve its performance. Fine-tuning the model on the target dataset led to a significant recovery in performance, with mAP50 and mAP50-95 improving (+39.4% and +31.5%), reaching 86.7% and 54.1%, respectively. This improvement was accompanied by a reduction in false negatives (-59%) and a recall increase (+45%), with the number of detected pomegranates rising from 1,753 to 5,653 out of 6,655 instances in the test set ([Fig. 10](#)).

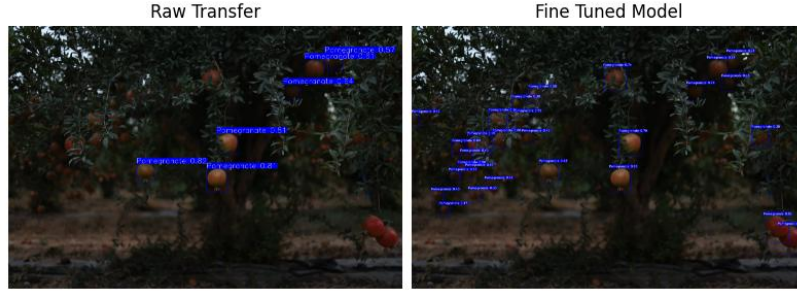


Fig. 10. (a) Raw Transfer and (b) Fine-tuned model pomegranate same image detection example.

Table 6

Object detection YOLOv8 different model test results.

Model	mAP50	mAP50-95	Precision	Recall	F1-Score
PG YOLO	0.938	0.597	0.881	0.868	0.870
Raw transfer	0.472	0.226	0.754	0.331	0.470
Fine-tuned model	0.866	0.541	0.860	0.780	0.820

4.3. Segmentation Unet Model Results

The U-Net segmentation model used in this study (section 3.3) employed an EfficientNet-B7 backbone and produced high-quality segmentation results, achieving an IoU of 84.5%, Precision of 89.1%, Recall of 78.7%, and an F1-Score of 83.6%. This configuration provided strong feature extraction capabilities and accurate pomegranate masks, supporting reliable downstream processing (Fig. 11). Segmentation outputs for all images in the dataset were generated using this model, and these predictions were used as input for the CNN+LSTM models.

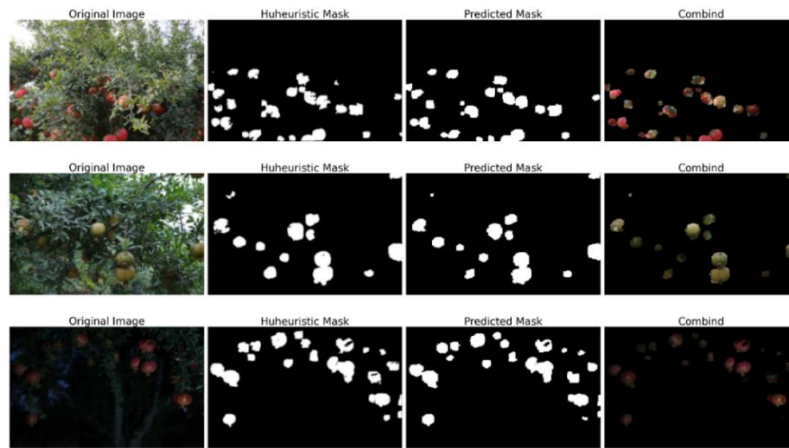


Fig. 11. Examples of test 3 model prediction results (a) original image (b) heuristic masks (c) prediction results (d) combined overlay mask and image.

4.4. CNN+LSTM tree cracking severity labeling model results

After the full optimization process combining the best batch size, learning rate, and image-processing configuration the final CNN+LSTM model achieved 68.8% accuracy, 55.7% AA, 56.3% precision, 45.9% recall, and 50.4% F1-score. The optimized setup used a batch size of 8, a learning

rate of 0.0001, and an enhanced image-processing pipeline consisting of CLAHE (clip limit = 2, tile size = 8×8), Gaussian blurring (kernel = 5), and Laplacian edge detection (kernel = 1). Compared to the same optimized model trained and tested only on the 2024 season, performance declined substantially across all evaluation metrics, with accuracy decreasing by -33.8%, AA by -44.4%, precision by -24.3%, recall by -14.3%, and the F1-score by -18.8%, indicating difficulty in maintaining robust performance under new seasonal conditions for this labeling scheme. Results are summarized in [Table 7](#).

Table 7
Summarized multiclass cracking severity labels results.

Dataset	Accuracy	AA	Precision	Recall	F1-Score
2024 (train/validation)	0.688	0.557	0.563	0.459	0.870
2025 (cross-season test)	0.350	0.113	0.320	0.316	0.316

4.5. CNN+LSTM tree fuzzy cracking severity labeling model results

The optimized fuzzy gaussian labeling configuration combined the best-performing settings from both batch-learning-rate and image-processing tuning. The final setup used a batch size of 8, a learning rate of 0.0001, and an image-processing pipeline consisting of CLAHE (clip limit = 9, grid size = 8), Gaussian blurring (kernel = 1), and Laplacian edge detection (kernel = 1). This configuration achieved the strongest fuzzy-label performance, producing the lowest MSE of 0.060, the lowest inter-side MSE of 0.007 and a high CA of 76.7%. Using the optimized model which was selected based on its performance when trained and tested only on the 2024 season, the fuzzy labeling scheme was further assessed by training on the 2024 dataset and testing on the 2025 season dataset, where the model demonstrated moderate generalization stability. Compared to the same optimized configuration trained and evaluated solely on the 2024 season, all fuzzy-based evaluation metrics exhibited changes across seasons. The MSE increased by +0.008, inter-side MSE increased by +0.01 and CA decreased by -8.7% indicating that the fuzzy labeling scheme maintained relatively stable performance under the new seasonal conditions. Results are summarized in [Table 8](#).

Table 8
Summarized fuzzy cracking severity labels results.

Dataset	MSE	Inter-Side MSE	Accuracy	CA
2024 (train/validation)	0.060	0.007	0.623	0.767
2025 (cross-season test)	0.068	0.017	0.460	0.680

4.6. CNN+LSTM tree binary cracking labeling model results

The optimized binary labeling configuration was built by adapting the CNN+LSTM architecture into a two-class setup to predict whether a tree would develop cracked fruit (1) or not (0). The final model used a batch size of 32, a learning rate of 0.001, and an image-processing pipeline consisting of CLAHE (clip limit = 3, grid size = 8), Gaussian blurring (kernel = 3), and Laplacian edge

detection (kernel = 3). This configuration achieved the strongest binary classification performance, reaching 84.1% accuracy, 70.4% AA, 75.3% precision, 78.3% recall, and a 76.6% F1-score, making it the most balanced and effective setup for predicting the presence or absence of fruit cracking. Using the optimized model which was first trained and tested on the 2024 season dataset using the binary cracking severity labels, the model was then evaluated on the 2025 season dataset to assess cross-season generalization. Compared to its performance when trained and tested solely on the 2024 season, the binary formulation exhibited clear reductions across all evaluation metrics, with accuracy decreasing by -19.1%, AA by -33.5%, precision by -29.8%, recall by -43.6%, and the F1-score by -37.2%, indicating limited generalization capability under the new seasonal conditions. Results are summarized in [Table 9](#).

Table 9

Summarized binary cracking labels results.

Dataset	Accuracy	AA	Precision	Recall	F1-Score
2024 (train/validation)	0.841	0.704	0.753	0.783	0.766
2025 (cross-season test)	0.650	0.369	0.455	0.347	0.394

4.7. Discussion

Predicting fruit cracking before visible symptoms appear is challenging due to strong variability in orchard conditions, sensor limitations, and the continuous nature of cracking progression. Illumination changes, occlusion, canopy structure, and inconsistent fruit visibility introduce noise that affects feature extraction and model stability. Seasonal shifts alter environmental conditions and fruit appearance, leading to noticeable changes in data patterns between years, which complicates the ability of models to generalize across seasons. Additionally, cracking progresses gradually rather than in discrete steps, making severity boundaries difficult to define and increasing the importance of selecting appropriate labeling strategies.

4.7.1. Detection and segmentation performance

The detection and segmentation stages, performed entirely on RGB images, established the foundation for the multimodal system. Fine-tuned YOLOv8 significantly improved over raw transfer, achieving 86.6% mAP50, 54.1% mAP50-95, 86.0% precision, 78.0% recall, and 82.0% F1-score, enabling reliable fruit localization under varied orchard conditions. Raw PG-YOLO transfer yielded 47.2% mAP50, 22.6% mAP50-95, 75.4% precision, 33.1% recall, and 47.0% F1-score, demonstrating the necessity of task-specific transfer learning to adapt the model to our orchard dataset. The U-Net segmentation model, using EfficientNet-B7, produced fruit masks with 84.5% IoU, 89.1% precision, 78.7% recall, and 83.6% F1-score, effectively isolating pomegranates and minimizing background noise. These results provided a modest and clean spatial input for the later cracking severity prediction stage.

4.7.2. Labeling schemes

Differences in labeling strategy had a clear impact on model performance. The optimized three-class CNN+LSTM model reached 68.8% accuracy, 55.7% AA, 56.3% precision, 45.9% recall, and 50.4%

F1-score, reflecting the difficulty of mapping a gradual cracking progression into fixed severity categories. The fuzzy-label formulation provided a smoother representation of this progression and achieved 0.060 MSE, 0.007 inter-side MSE and CA of 76.7% indicating more stable and overall improved performance compared with the multi-class formulation. The binary labeling setup produced the strongest supervised results, reaching 84.1% accuracy, 70.4% AA, 75.3% precision, 78.3% recall, and 76.6% F1-score, making it the most effective labeling scheme strategy for predicting cracking and whether cracking would occur.

4.7.3. Generalization across seasons

Cross-season evaluation (train-2024 / test-2025) showed clear differences in how well each labeling scheme retained performance when applied to new seasonal conditions. Generalization was evaluated by examining the deviation of each model's 2025 results from the expected behavior of its optimized configuration. The three-class model showed the largest degradation. In 2025 it achieved 35.0% accuracy, 11.3% AA, 32.0% precision, 31.6% recall, and an F1-score of 31.6%, indicating that the original hard labeling scheme did not transfer reliably across seasons, suggesting that the model struggled to capture stable patterns in cracking severity under varying environmental and visual conditions. The sharp reductions across all evaluation metrics show that this label structure was not robust for year-to-year prediction, limiting the model's ability to generalize beyond the conditions present in the training season. The binary model maintained performance more effectively. On the 2025 dataset it reached 65.0% accuracy, 36.9% AA, 61.4% precision, 66.2% recall, and an F1-score of 39.4%, demonstrating that the binary labeling scheme showed limited generalization capability, with accuracy remaining only moderately stable while AA and the other evaluation metrics dropped sharply. The much larger reduction in AA relative to accuracy suggests that cross-side consistency deteriorated considerably under the 2025 conditions, highlighting that the model struggled to maintain reliable and aligned predictions between the two sides of the field when transferred to a new season. The gaussian fuzzy-label model demonstrated the strongest generalization performance, reaching 0.068 MSE, 0.017 inter-side MSE and 68.0% CA on the 2025 dataset, reflecting the smallest performance degradation across seasons and the highest CA relative to the accuracies achieved by the other two labeling schemes, suggesting that the fuzzy label structure supports improved generalization and more stable prediction behavior under new seasonal conditions.

4.7.4. Advantages and limitations

This study showed that the choice of labeling scheme played a central role in modeling pomegranate cracking. The predicted labels at the tree level reflected how each approach three-class, fuzzy, and binary captured differences in cracking severity, with the strategies varying in how clearly, they separated trees with distinct risk levels. Cross-season evaluation further demonstrated that these labeling schemes responded differently to changes in environmental conditions, with each showing a different level of robustness when applied to a new season. The dataset covered a single orchard and only two seasons, limiting the representation of different environments, cultivars, and growth conditions. Class imbalance made multi-class prediction difficult, especially for borderline cracking cases where visual differences are subtle. Misalignment between thermal and RGB cameras prevented pixel-level fusion, requiring thermal information to be used only as tabular features. Variability in acquisition such as inconsistent viewpoints, occlusions, and occasional tree-

identification errors introduced additional noise into the temporal sequences. Future work should expand data collection across additional trees, rows, and seasons, and improve acquisition consistency to reduce noise. Better thermal-RGB calibration would enable pixel-level multimodal fusion, and extending the time series may help capture early physiological signals that precede cracking. Exploring hybrid labeling strategies that combine the interpretability of binary labels with the robustness of fuzzy representations may help balance interpretability and stability in future models.

5. Conclusions

This study developed a multimodal deep learning framework for predicting on-tree pomegranate fruit cracking by integrating RGB imagery, thermal features, and meteorological measurements. The 2024 dataset included 9,276 RGB images forming 3,092 temporal sequences aligned across sensing modalities, and the 2025 dataset included 3,024 images forming 1,008 sequences for cross-season evaluation. Core components of the framework included fruit detection using YOLOv8, which achieved strong performance with an mAP50 of 86.6% and an mAP50-95 of 54.1%, and semantic segmentation using U-Net with grab cut heuristic masks, which reached an IoU of 84.5%. Three labeling strategies were evaluated, and the results showed clear differences in predictive behavior. The binary labeling scheme provided the strongest supervised performance, achieving 84.1% accuracy and 70.4% AA, making it the effective framework for early-warning prediction of cracking occurrence. In contrast, the fuzzy labeling scheme demonstrated the highest stability and the best cross-season generalization. The Gaussian fuzzy formulation reached 0.060 MSE, 0.829 cosine similarity, and 0.007 inter-side MSE, offering a smooth representation of the gradual cracking process. When evaluated under a train-2024/test-2025 setup, fuzzy labels showed the smallest performance change, achieving 0.068 MSE, 0.017 inter-side MSE and 68.0% CA. These results confirm that fuzzy labeling is more resilient to seasonal variation, illumination changes, and tree-level shifts than discrete labels. Overall, this study shows that multimodal sensing paired with appropriate label design can effectively support pomegranate cracking prediction at the tree level. Binary labeling provides strong supervised accuracy for decision-making, while gaussian fuzzy labeling offers superior stability across seasons, underscoring the importance of choosing a labeling scheme that remains reliable under varying field conditions.

CRedit authorship contribution statement

Arad Peleg: Conceptualization, Methodology, Software, Investigation, Data curation (2024 dataset), Formal analysis, Visualization, Writing - original draft. **Victor Alchanatis:** Supervision, Investigation, Data curation (2025 dataset), Writing - review & editing. **Yael Edan:** Supervision, Project administration, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The 2024 dataset used in this study can be downloaded from the following link: https://docs.google.com/document/d/1_rmgK5q6M9Lcu5B8nKeYlqPtpLuIcN6V/edit?usp=drive_lin

[k&oid=100602072493698229200&rtpof=true&sd=true](https://doi.org/10.1007/S11119-023-10084-Y/TABLES/5). The same link also contains the 2025 dataset, however, access to the 2025 data requires approval from the dataset owner.

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